

Muhammad Yahya

Email: yahyasid45@gmail.com | Phone: +92 316 3434749 | Hyderabad, Sindh, Pakistan
LinkedIn: linkedin.com/in/muhammad-yahya-siddiqui-97967327a
GitHub: github.com/yahyaCodeX

Professional Summary

Computer System Engineering student with practical experience in Java backend development, Spring Boot, microservices architecture, blockchain-integrated systems, and RESTful API development. Skilled in scalable backend engineering, Apache Kafka, JWT/OAuth2 authentication, Docker, MySQL, PostgreSQL, and AI integrations using OpenAI APIs. Passionate about distributed systems, secure application development, and cloud-native technologies.

Education

Mehran University of Engineering and Technology (MUET) — Computer System Engineering
Jamshoro, Sindh | Dec 2022 – Dec 2026
CGPA: 3.22/4.00

Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Software Engineering

Technical Skills

- **Programming Languages:** Java, SQL
- **Frameworks & Tools:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, Docker
- **Microservices & APIs:** REST APIs, Spring Cloud, Apache Kafka, API Gateway, Config Server
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Blockchain:** Ganache, IPFS, Smart Contract Integration, Blockchain-Based Verification Systems
- **Security:** JWT Authentication, OAuth2, Role-Based Access Control (RBAC), Keycloak
- **AI & Machine Learning:** LLM Integration, Prompt Engineering, OpenAI API
- **Version Control:** Git, GitHub

Projects

Microservices-Based Fitness Tracker

Java, Spring Boot, Apache Kafka, Keycloak, PostgreSQL, Docker

- Architected a scalable microservices ecosystem using Spring Cloud and Apache Kafka for real-time activity event streaming.
- Implemented OAuth2/OpenID Connect authentication and authorization using Keycloak.
- Developed RESTful APIs and distributed backend services for fitness activity and user management.
- Containerized services using Docker for deployment consistency and scalability.

Next-Generation Library Management System

Java, Spring Boot, Stripe API, Hibernate/JPA, MySQL

- Integrated Stripe Payment Gateway for automated fee collection and membership subscription handling.
- Designed and implemented secure backend APIs for inventory, transactions, and user management.
- Optimized database operations using Hibernate/JPA for efficient data persistence.

Blockchain-Based Degree Verification System

Java, Spring Boot, MySQL, Ganache, IPFS, JWT Authentication

- Developed a secure blockchain-integrated degree verification platform for validating academic credentials.
- Integrated IPFS for decentralized storage of degree documents and Ganache for blockchain transaction simulation.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) for secure system access.
- Built RESTful APIs using Spring Boot for certificate issuance, verification, and user management.
- Designed MySQL database schemas for managing academic records and verification workflows.

Professional Experience

Web Development Intern

Pakistan Engineering Forum (PEF) – MUET Chapter | 2025 | 6 Weeks

- Collaborated with a development team to maintain and improve the organization's web presence.
- Gained practical exposure to the Software Development Life Cycle (SDLC) in a professional engineering environment.
- Contributed to frontend and backend web development tasks while improving teamwork and communication skills.

Certifications

- Introduction to Artificial Intelligence — Coursera
- Graphic Design Certification — National Freelance Training Program (NFTP)